



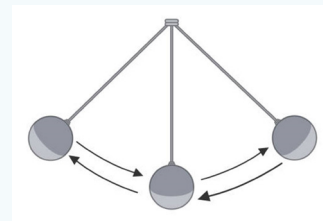
We observe different motions around us. One such motion can help us to keep track of time using the pendulum. Let's create an animation of pendulum.



Animation has become a big part of our lives. It is used everywhere; in movies, games, cartoons, in the education industry, to visualise experiments, etc. Animation is used to show imaginary events or ones that we cannot easily see. In this lesson, we will look at simple animations and how to create them.

Animation can be used to make an object move or change shape over time. **Animations** are made with Computer-Generated Imagery (CGI). Let's make an animation of a pendulum.

First, we need to know a little more about pendulums. A pendulum is a weight suspended from a pivot so that it can swing freely. A pendulum is used to regulate the movement of clocks. The time interval for each complete oscillation is constant. A complete oscillation is also called a period.



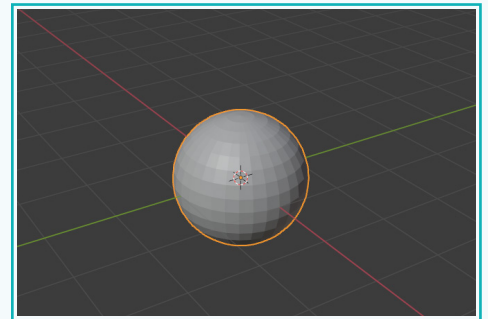
We will create an animation in Blender to show a pendulum's movement.

Activity

- Use Blender to create an animated pendulum.

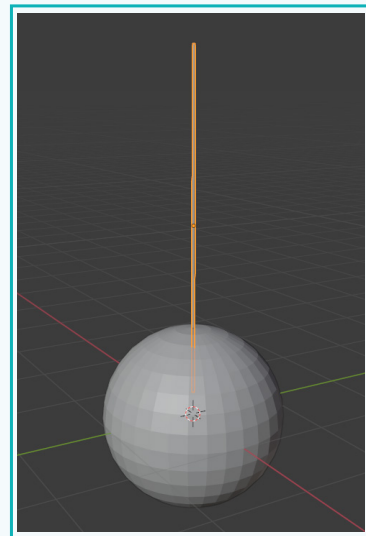
To create this animation, we will start with creating a model.

- 1 Create a UV sphere mesh.



- 2 Create the upper part of the pendulum.

- Create the upper part of the pendulum.



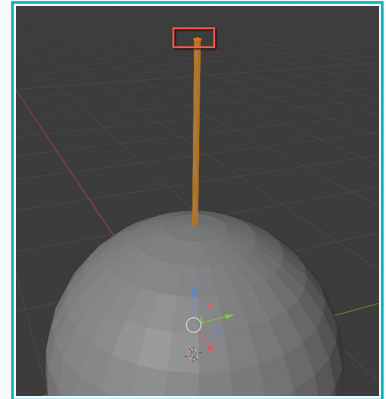
A pendulum oscillates from its topmost part. Currently, the object is moving along the centre of the mesh.

Bytes

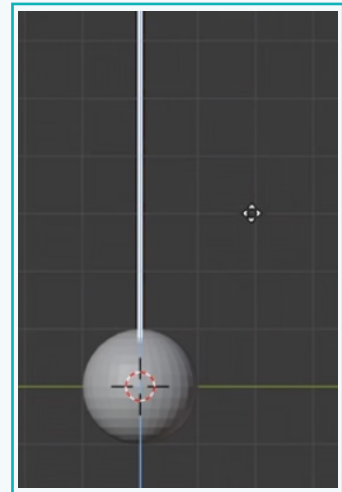
Galileo, an Italian scientist, first observed the constant movement of a pendulum's period. He compared the movement of a swinging lamp in cathedral with his pulse rate. In 1656 a Dutch mathematician and scientist, Christiaan Huygens, invented a clock controlled by the motion of a pendulum. French physicists assembled large pendulums called Foucault pendulums and performed the first laboratory demonstration of the Earth's rotation of Earth around its axis.

Activity

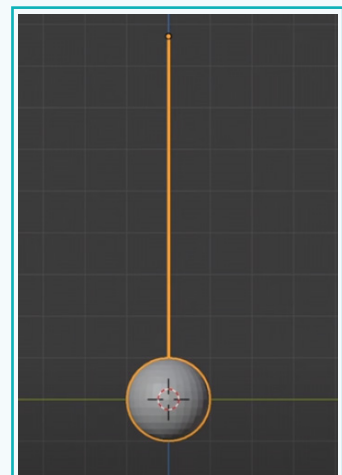
- 3 Change the centre point.
 - a. Press the tab key to switch to edit mode.
 - b. Press A to select all the faces. Lower them along the z axis.
 - c. Make sure the all faces are below the centre point.
 - d. A key toggles selection of all mesh components/objects.
 - e. Press tab to switch to object mode.



- 4 Place the objects in their correct positions.



- 5 Join the objects.
 - Once the objects are placed, select the UV sphere. Then, select the cylinder while pressing shift. Use Ctrl + J to join the objects. Now we will be able to animate all the parts of the pendulum as one object.



Activity

Let's start to animate!

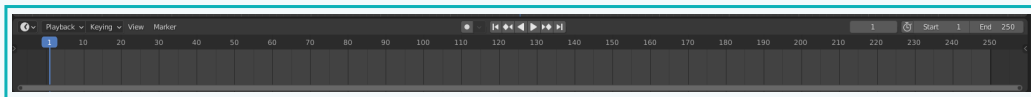
Animation makes an object move or change shape over time. This is typically achieved with the use of keyframes. A keyframe is a location on a timeline which marks the beginning or end of a transition. The purpose of a Keyframe is to allow for interpolated animation.

For example, a user can add a key on frame 1 and another key on frame 120, which specify the pendulum's angles of rotation. Blender will automatically determine the correct rotation of the pendulum for all the frames between frame 1 and 120.

Note:

The **angle of rotation** is a measurement of the amount that a figure is rotated about a fixed point.

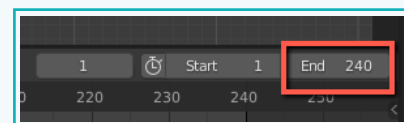
Keyframes are set over the timeline panel, which is visible below the viewport.



The Timeline editor, identified by a clock icon, is used to manipulate keyframes.

The timeline gives the user a broad overview of a scene's animation. It displays the current frame, the keyframes of the active object, the start and end frames of an animation sequence, as well as markers set by the user.

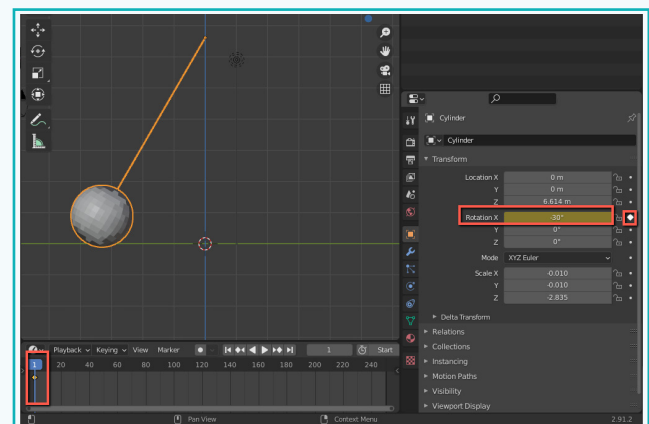
Set the end frame as 240.



The timeline includes Transport Controls to play, pause, and skip through an animation sequence. It also includes tools to manipulate keyframes, keying sets, and markers.

6 Set the rotation Angle.

- For our pendulum to oscillate, we shall define the angle of rotation along one single axis.
- Select the pendulum, making sure that we are at frame 1. Set the we angle of rotation along the x axis as -30.



Activity

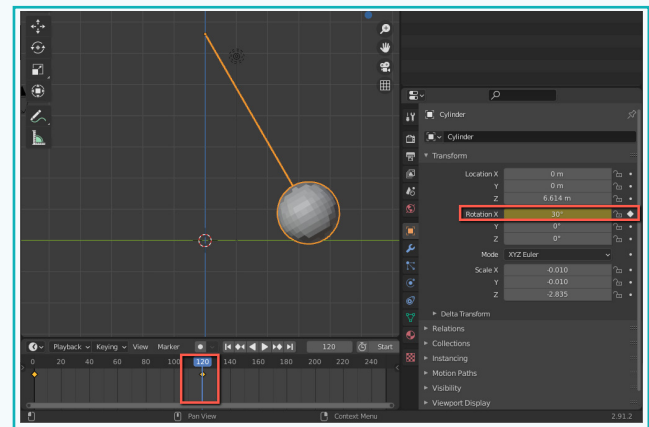
- c. To set the keyframe, press the key button besides the rotation X value.

Observe that the key turns to a diamond shape. On the timeline we can see a yellow diamond at frame 1. The number of frames that are played in 1 second is called the frame rate.

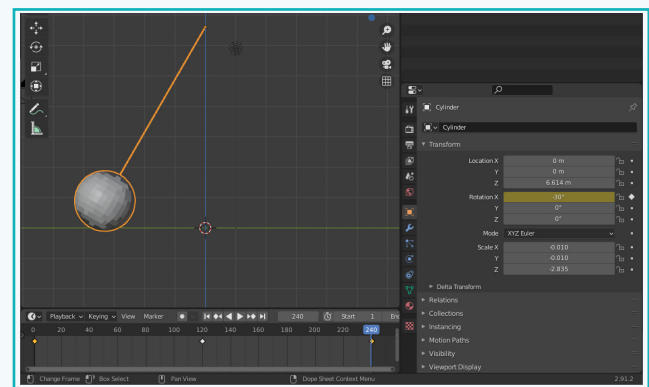
Frames Per Second (FPS) is the unit used to measure frame rate. A human eye can process 16 separate images per second. This means a frame rate of 16 FPS can display motion, but the animation will appear abrupt and disconnected. Once the frame rate exceeds 16 FPS, the frames appear less discrete and start to blur into each other. A frame rate of 24 FPS is set by default in Blender to create smooth transitions.

- 7 Set another keyframe.

- a. Set another keyframe for the pendulum to oscillate towards the other side of the axis.



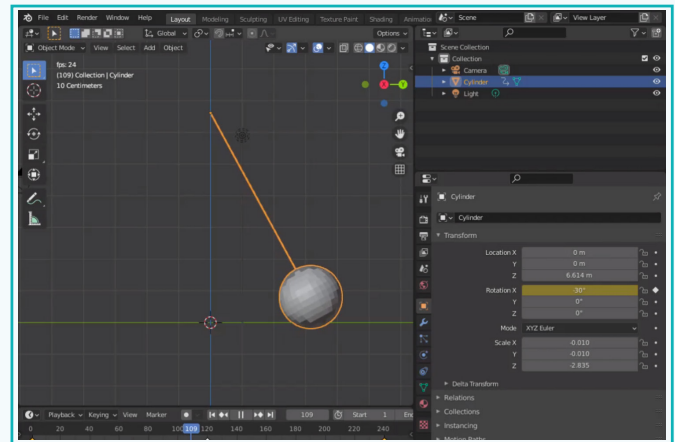
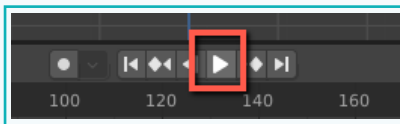
- b. Similarly, we will set the pendulum to swing back to the starting position at frame 240.



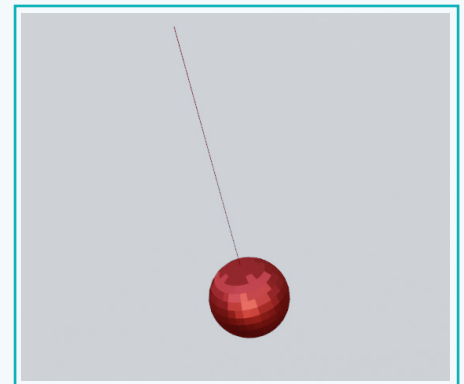
Here, we have successfully set the frames for one complete oscillation of the pendulum.

Activity

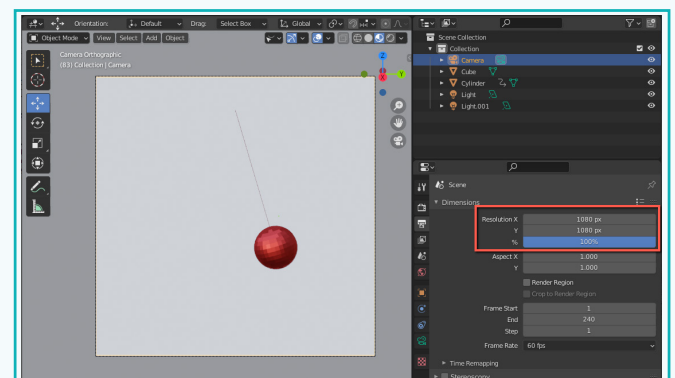
To view the animation, press the play button in the timeline transport controls.



Now that animation is ready, we will add other elements and materials to complete the scene.



Before we export the animated video, we need to correctly set the camera angle. Tumble around the viewport to find the desired view. To set the camera at the viewing angle, press Ctrl+Alt+0. We can also set the render resolution in the scene properties.

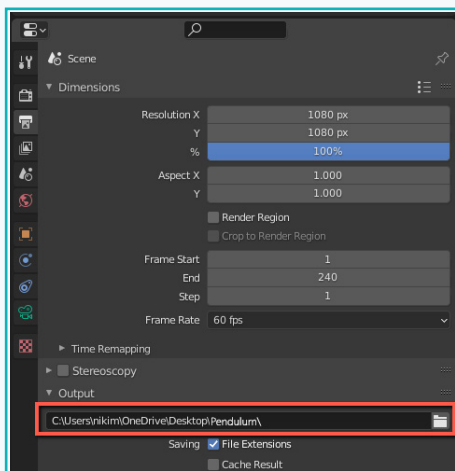


Rendering is the process of generating an image from a 3D model using a computer application.

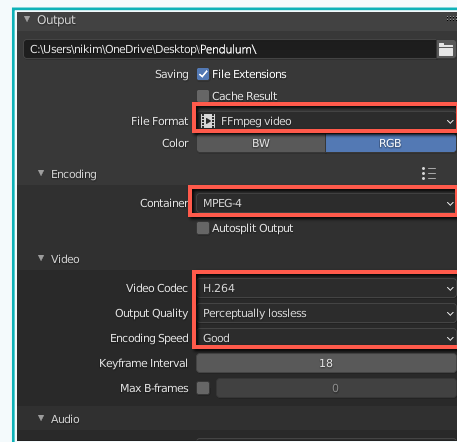
Activity

8 Render the output.

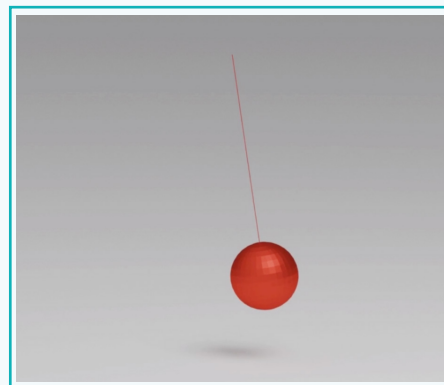
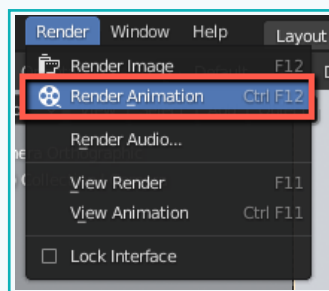
a. Set the output directory where our rendered file will be stored.



b. Set its format according to the image below.



a. Now that all the settings have been corrected, we are ready to render the animation.



In this lesson, we learnt about the need for animation. We also found out some information about the pendulum. We understood how keyframes work, and then we rendered and created our own animated pendulum.

Exercise

Fill in the blanks

- 1 _____ are set over the timeline panel.
- 2 The timeline editor is identified by a _____ icon.
- 3 The timeline includes _____ to play, pause, and skip through an animation sequence.
- 4 FPS is the unit used to measure the _____.

See how easy it was. Now, you can animate events that are not visible to us; such as, solar system or germination of seed.



Tech Flash

- **Animation** : Animation is a process to make object move or change shape over time.
- **Keyframe** : A keyframe is a location on a timeline which marks the beginning or end of a transition.
- **Rendering** : Rendering is the process of generating an image from a 3D model using a computer application.

- **Hot keys** :

A	Toggle selection of all mesh components/objects
Ctrl + J	Join objects
Ctrl+Alt+0	Move the active camera to view
0	View active camera